

Public Grievance Urgency Classification System

AI-Powered Triage and Response Optimization for Civic Grievances

Track: AIML & Civic Operations

Focus: Automatic Routing, Priority Tagging, & Escalation Recommendations

The Challenge in Public Service

Current Backlogs: Public grievance systems receive hundreds of thousands of citizen complaints weekly, causing severe operational delays.

Manual Bottlenecks: Human operators spend hours categorizing, checking, and manually routing files to departments, delaying responses.

Critical Issue Delay: Life-critical matters (e.g. water cuts, power outages, public safety hazards) get lost in routine queues.

Goal: Build a text classification system that detects category and urgency level so serious issues are routed and resolved in under 24 hours.

Dataset & Feature Extraction

Primary Source: Government of India Grievance Report.

Input Data: Extracted and cleaned text from `subject_content_text` and `remarks_text`.

Output Targets:

- **Category Description:** Decoded from the CategoryV7 mapping index (e.g., Railways, Defence, Rural Development, Public Health).
- **Urgency Score:** Rule-based scoring (High, Medium, Low) using NLP keywords to prioritize critical terms.

System Architecture & Workflow

The end-to-end pipeline consists of the following components:

1. **Data Pipeline:** Reads the raw dataset from JSON, cleans MongoDB attributes, and processes text fields.
2. **EDA Visuals:** Analyzes category frequency distributions and regional volume counts.
3. **Machine Learning Model:** TF-IDF vectorization coupled with a Calibrated Logistic Regression classifier.
4. **Actionable Outputs:** Combines predictions and confidence scores to output routing and resolution timelines (SLAs).

Model Optimization & Features

Feature Engineering: Word n-grams (1 to 3) + Character n-grams (3 to 5) combined via a Feature Union to capture spelling variations and semantic context.

Classification Algorithm: Logistic Regression with balanced class weights to manage category imbalances.

Probability Calibration: CalibratedClassifierCV wrapper is applied to generate reliable confidence probability outputs.

NLP Priority Override: Live checks for safety-critical terms automatically upgrade complaints to High Urgency.

Prototype Deployment

FastAPI Microservice (Backend):

- Exposes a /predict POST endpoint receiving raw text payloads.
- Performs real-time preprocessing, category prediction, and urgency classification.
- Returns structured JSON including confidence metrics and SLA recommendations.

Streamlit Dashboard (Frontend):

- Accessible, simple text area for operators or citizens to enter grievances.
- Displays predictions in color-coded metrics alongside clear actionable recommendations.

Validation and Performance

High Accuracy: The model achieves ~98.7% test accuracy on the top 10 categories, verifying robustness.

Confidence Scoring: Calibrated probabilities enable filtering out predictions below 70% confidence for manual oversight.

Response SLA Metrics:

- **High Urgency:** Target resolution within 24 hours (Nodal Officer Escalation).
- **Medium Urgency:** Target resolution within 3 days (Department Lead).
- **Low Urgency:** Target resolution within 7 days (Routine flow).

Current Limitations

- Static Heuristics:** Urgency tags rely on keyword matches. Complex sentiment or contextual sarcasm may not be correctly classified.
- Imbalanced Coverage:** Very rare categories are filtered out during model training to ensure prediction robustness, leaving them for manual routing.
- Hardware Dependency:** The tf-idf n-gram vocabulary is large (~80,000 features), leading to memory overhead for real-time edge devices.

Next Steps & Roadmap

Deep Learning & LLMs: Upgrade keyword urgency mapping to semantic embedding classifiers (e.g. DistilBERT) to detect nuanced urgency.

Automated Action History: Incorporate citizen response feedback from action logs to iteratively retrain model weights.

Geo-Aware Routing: Leverage state and district tags to auto-dispatch files directly to local municipal ward offices.

Multi-Lingual Support: Extend cleaning heuristics to Hindi (Devanagari script) and regional languages to improve accessibility.